

Hidden Follower Detection via Refined Gaze and Walking State Estimation

Yaxi Chen^{a,b}, Ruimin Hu^{a,b(✉)}, Danni Xu^c, Zheng Wang^{a,b}, Linbo Luo^d, Dengshi Li^{a,e}

^a National Engineering Research Center for Multimedia Software, School of Computer Science, Wuhan University, Wuhan, China

^b Hubei Key Laboratory of Multimedia and Network Communication Engineering, Wuhan University, Wuhan, China

^c School of Computing, the National University of Singapore, Singapore

^d School of Cyber Engineering, Xidian University, Xi'an, China

^e School of Artificial Intelligence, Jiangnan University, Wuhan, China

{chenyaxi, hrm}@whu.edu.cn, dannixu.gracie@foxmail.com, wangzwhu@whu.edu.cn, lbluo@xidian.edu.cn, reallds@jhu.edu.cn

Abstract—Hidden following is following behavior with special intentions, and detecting hidden following behavior can prevent many criminal activities in advance. The previous method uses gaze and spacing behaviors to distinguish hidden followers from normal pedestrians. However, they express gaze behaviors in a coarse-grained way with binary values, making it difficult to accurately depict the gaze state of pedestrians. To this end, we propose the Refined Hidden Follower Detection (RHFD) model by choosing a suitable mapping function based on the principle that the closer the gaze direction is to someone, the more likely it is to gaze at someone, which converts the gaze direction into a continuous estimated gaze state representing the complex and variable gaze behavior of pedestrians. Simultaneously, we introduce variations in the magnitude and direction of pedestrian velocity to refine the representation of pedestrian walking states. Experimental results on the surveillance dataset show that RHFD outperforms state-of-the-art methods.

Index Terms—hidden follower detection, human activity recognition, social science

I. INTRODUCTION

Human behavior recognition is currently the main research hotspot in video understanding [1]–[6], which has potentially great value in the field of surveillance and security. Hidden following is a kind of following behavior with special hidden intentions with characteristics such as concealment, and detecting it also depends on the understanding of human behaviors. It is not an ongoing criminal activity, but a preparatory activity for many illegal and criminal acts against national public security. Accordingly, timely detection of such hidden abnormal behavior can alert a series of criminal events, such as terrorism.

In recent years, most traditional following detection methods [7], [8] are mainly based on the trajectory to determine whether the pedestrian is following the target object. However, in public places where people are walking very closed and many people are on the same path, trajectory similarity happens from time to time, which makes it impossible to accurately detect hidden and really dangerous following behaviors. To overcome this defect, the recent work [9] proposed to use gaze and spacing flow to recognize hidden followers, however, the method has the following limitations: 1) There

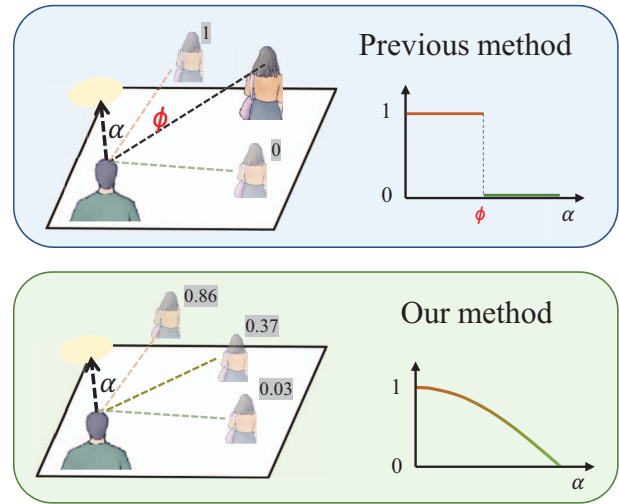


Fig. 1. The previous method uses a coarse-grained representation of threshold dichotomy for gaze behavior and lacks a refined representation of gaze state. In this paper, we express the gaze behavior of pedestrians by using a continuous estimated gaze state, to further consider the complex variability of follower gaze states, where α denotes the gaze direction and ϕ denotes the threshold angle degree.

is a confidence that the binary gaze value is wrongly estimated, thus, using a coarse-grained representation of threshold dichotomy for gaze behavior cannot realistically and effectively characterize the refined gaze modes of pedestrians. 2) In the following process, there are differences in velocity magnitude and direction between hidden followers and normal pedestrians, and without considering the walking velocities of pedestrians, the variations in the walking states of pedestrians cannot be fully characterized.

The challenges of hidden follower detection are mainly in the following: 1) **Hidden Followers**. Hidden followers have to hide their following behavior, they do not have obvious physical postures and actions during the following process, and it lasts for a long time. In addition, the behavioral mode of hidden followers may vary widely among individuals due

to individual differences. 2) **Following Scene.** The following scene is complex and variable, and it is difficult for the general security department to directly obtain the precise daily walking trajectories of all pedestrians. Secondly, the hidden following mode of the same individual varies greatly due to environmental factors, such as pedestrian density, and road complexity.

Our main idea is that the complex and variable gaze behaviors of hidden followers under continuous time lead to continuous variations in their gaze state. In general, it is hard to determine whether a person is gazing at another person. The gaze state between "gaze" and "not gaze" does not change abruptly either. Therefore, the estimated gaze state of a pedestrian at each moment in the same surveillance video is probabilistic and continuous. In addition, in the task of the hidden following, the hidden follower prevents losing the target pedestrian, and the magnitude and direction of his velocity will be as similar as possible to that of the target pedestrian and will differ significantly from that of the normal pedestrian.

To this end, we design a Refined Hidden Follower Detection (RHFD) framework, which not only can refinedly characterize the complex gaze modes of pedestrians but also precisely capture the behavior of hidden following in the real world. Specifically, our model consists of three main components, namely the gaze mode module, the velocity mode module, and the spacing mode module. The gaze mode module is responsible for refining the representation of the continuous estimated gaze states of pedestrians, using historically estimated gaze states to improve the currently estimated gaze, increasing the chance of capturing gaze features, while the velocity mode module is responsible for refinedly expressing the walking state of the pedestrians, using the difference between the magnitude and direction of the velocity of hidden followers and normal pedestrians. We validate the effectiveness of RHFD under the surveillance dataset. The experimental results show that RHFD outperforms state-of-the-art methods in the hidden follower detection task.

The contributions of our paper are summarized as follows:

- We propose a new perspective on understanding and expressing pedestrian gaze behavior and walking states under surveillance video. Unlike existing methods, we consider the continuous estimated gaze state of the pedestrian and the velocity magnitude and direction of the pedestrian walk in continuous time.
- We explore the behavioral relationship between hidden followers and normal pedestrians, discovered the trend of estimated gaze state changes of hidden followers during walking, and verified that the estimated gaze state and velocity characteristics of hidden and normal pedestrians were significantly different.
- We construct a novel refined hidden follower detection algorithm and experimentally validate its improvement in several metrics compared to the benchmark methods.

II. RELATED WORK

Hidden follower detection is an area of research that has emerged in recent years. Since the field was first proposed in 2021 [10], major research has focused on mining the differences between hidden and non-hidden followers through gaze patterns and spacing patterns to design a framework for hidden follower detection. Although these methods can address the shortcomings of the traditional following detection methods to some extent, it still suffers from the incomplete and ambiguous expression of behavioral patterns. Therefore, to better identify hidden followers and accelerate the advancement of this research area, there is an urgent need to capture pedestrians' behavioral patterns comprehensively from different dimensions.

To sum up, our RHFD model differs from other models in the following points: 1) Our study considers the complex and variable gaze state of hidden followers and shift the gaze behavior into a continuously variable estimated gaze state. 2) We find differences in the magnitude and direction of the walking velocity of hidden followers and normal pedestrians.

III. PROPOSED METHOD

A. Problem Formulation

Given a target pedestrian M and a surveillance video S containing M , the set of pedestrians in the video is $\mathbf{X}$, $x \in \mathbf{X}$, and the set of hidden followers is $\mathbf{F}$. If the probability of being recognized as a hidden follower is $p_x > 0.5$, then $x \in \mathbf{F}$.

Estimated Gaze State. The extent to which the hidden follower gazes at the target pedestrian over a continuous time.

B. RHFD Framework Overview

We employ a Refined Hidden Follower Detection (RHFD) model that considers continuously variable gaze states and walking velocities of pedestrians. We use estimated gaze state frequency (Gf) and density (Gd) for gaze modes, velocity difference (Vv) and direction (Vd) for velocity modes, and distance range (Dr) and mean distance (Dm) for spacing modes. The spacing mode module is the same as [9]. As shown in Fig.2, the framework first extracts the trajectory of pedestrians from the video by using the TraDes [11] and obtains the unconstrained gaze direction by using the Gaze360 [12], then we obtain the gaze mode, velocity mode, and spacing mode, finally, input them to the time series classification model to recognize the hidden followers. In what follows, we provide detailed descriptions of the gaze mode module, the velocity mode module, the spacing mode module, and the time series classification.

1) *Gaze Mode Module:* We extract the unconstrained gaze direction of each pedestrian from the gaze360 [12], and derive the gaze direction α from the position coordinates of the pedestrian and the unconstrained gaze direction, as follows [9]:

$$\Delta x = x_p - x_m, \Delta y = y_p - y_m, \quad (1)$$

$$\alpha = \arccos \frac{(d_g^x, d_g^y) \cdot (\Delta x, \Delta y)}{|(d_g^x, d_g^y)| \cdot |\Delta x, \Delta y|}, \quad (2)$$

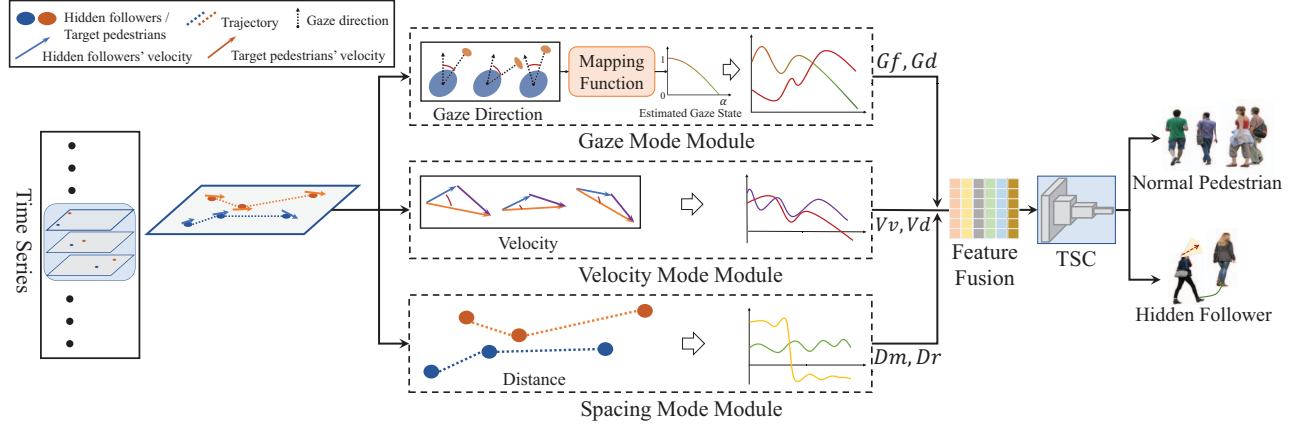


Fig. 2. The framework of the RHFD model. It consists of a gaze mode module, a velocity mode module, and a spacing mode module. Firstly, the trajectory and unconstrained gaze direction of the pedestrian in the video are extracted, then the modes of estimated gaze state, velocity, and spacing of the pedestrian are represented respectively, and finally, they are fed to the time series classification model to recognize the hidden followers.

where (x_p, y_p) and (x_m, y_m) are the ground coordinates of P and M , (d_g^x, d_g^y) is the 2D gaze direction of P .

In general, the smaller the gaze direction α of P , the greater the probability that P is gazing at M . In previous work [9], the relationship between α and the magnitude of the set threshold was used to define the gaze state g . Such a direct comparison with the given threshold to determine whether it is gaze or not is prone to edge blurring near the threshold, and secondly, because of the inconsistency between the direction of gaze through the eye and the head position, the gaze direction obtained by model extraction will have some error, and the direct adoption of the threshold judgment method does not truly and effectively characterize the complex and variable gaze mode of the pedestrian. Therefore, this method converts the uncertainty in the presence of errors into probabilistic events by using the mapping relationship to convert the gaze direction into a continuous estimated gaze state as follows:

$$g = \begin{cases} \mathcal{F}(\alpha), & \alpha \leq \phi \\ 0, & \alpha > \phi \end{cases}, \quad (3)$$

where $\mathcal{F}(\alpha)$ is the mapping function, ϕ is set to 90 degrees.

There is no evidence of a clear relationship between the gaze direction and gaze state of hidden followers. Therefore, this paper proposes a mapping function $y = f(x)$, which conforms to two general rules: the closer the gaze direction is to someone, the more likely it is to be a gaze. Mathematically, the smaller x is, the larger y is. Also, since the gaze direction goes from 0 to 90 degrees, the probability of gaze should decrease from 1 to 0. This paper uses the normal assumption that the relationship between gaze direction and gaze state satisfies a polynomial function based on the above rules. To simulate and discover the relationship between gaze direction and the estimated gaze state, and to detect hidden followers, this paper initially tried four different functions, as shown in Fig.3, that have the same gaze direction with different

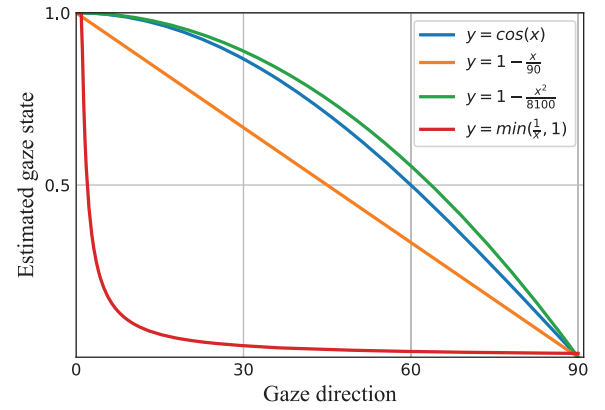


Fig. 3. The trend of conversion of gaze direction to estimated gaze state under different mapping functions.

estimated gaze states. The mapping functions are denoted as follows:

$$y_1 = \cos(x), y_2 = 1 - \frac{x}{90}, y_3 = 1 - \frac{x^2}{8100}, y_4 = \min(\frac{1}{x}, 1). \quad (4)$$

Moreover, it is not precise enough to determine the extent of the gaze state of a hidden follower P based on the gaze direction at a moment in time, because gaze should be a continuously changing process. Therefore, we added exponential averaging to link the estimated gaze state g_t at a moment t to the historical estimated gaze state, increasing the chance of capturing historical gaze characteristics, as follows:

$$g_t = \frac{g_t + (1 - \beta)g_{t-1} + (1 - \beta)^2g_{t-2} + \dots + (1 - \beta)^tg_0}{1 + (1 - \beta) + (1 - \beta)^2 + \dots + (1 - \beta)^t}, \quad (5)$$

where β is the smoothing coefficient, which is set to 0.95 in this work.

We calculate the frequency (Gf) and the density (Gd) of the estimated gaze state: Given an L -second video, we divide it into several K -second video clips. Each clip contains $R \times K$ frames, where R represents the frame rate. As follow [9]:

$$Gf = \frac{\sum_{t=2}^{R \times K} |g_t - g_{t-1}|}{R \times K}, Gd = \frac{\sum g_t}{R \times K}. \quad (6)$$

2) *Velocity Mode Module*: The walking velocity of a pedestrian in a hidden following task is an important indicator to determine whether there is a hidden following relationship. For a hidden follower, its intention to follow the target object determines the direction and magnitude of its speed, which cannot deviate too much from the target object. The hidden follower may control the position relationship with the target object by controlling the velocity and direction, accelerating and following in a straight line when it is easy to lose the target object, and slowing down or deviating from the direction when it is easy to be detected by the target.

To this end, this paper proposes a hypothesis that hidden followers differ significantly from normal pedestrians in terms of velocity characteristics, which is verified by the KS test. Second, although gaze behavior is an effective feature for identifying hidden followers, gaze behavior itself is difficult to estimate accurately in videos taken at medium and long distances. In contrast, the velocity feature is less dependent on the distance and quality of the video shot, which makes it an important attribute. Combined with the gaze feature, the velocity feature can obtain a more stable and accurate recognition performance. Therefore, the velocity feature is introduced in this paper to improve the accuracy of the model.

To extract the velocity information of the pedestrian, the pixel coordinates of the pedestrian in the image are first extracted using the TraDes model [11]. Then a perspective transformation matrix is calculated by corresponding the pixel coordinates of the four points in the image to the ground coordinates in the top view. Using this perspective transformation matrix, the pixel coordinates of the pedestrian can be transformed into the two-dimensional position coordinates in the top view. Suppose the position coordinates of a pedestrian at the moment t are denoted by (x_t, y_t) , then the velocity of the pedestrian at that moment can be coordinated as:

$$\vec{v}^t = \left(\frac{x^{t+1} - x^{t-1}}{2}, \frac{y^{t+1} - y^{t-1}}{2} \right). \quad (7)$$

The relative velocity and the cosine of the direction of the velocity of P and M at any moment can be expressed as:

$$Vv = \bar{v} = |\vec{v}_p - \vec{v}_m|, Vd = \cos \theta = \frac{\vec{v}_p \cdot \vec{v}_m}{|\vec{v}_p| \cdot |\vec{v}_m|}, \quad (8)$$

where $|\cdot|$ denotes the norm of the vector.

3) *Spacing Mode Module*: As with the velocity mode module, we first obtain the 2D position coordinates in the top view of the pedestrian. Given a L -second video, it is divided into several K -second video clips. Each segment contains $R \times K$ frames, where R represents the frame rate. The distance of each pair of pedestrians is calculated from the 2D position

coordinates of the overhead view of the pedestrians, and the distance sequence $\mathcal{D} = \{d_1, d_2, \dots, d_{R \times K}\}$ is obtained, and the distance range (Dr) and the average distance (Dm) can be expressed as:

$$Dr = \max(\mathcal{D}) - \min(\mathcal{D}), Dm = \sum_{i=1}^{R \times K} \frac{d_i}{R \times K}. \quad (9)$$

4) *Time Series Classification*: To be able to accurately identify hidden followers, this study invokes the time series classification model, TWIESNs [13]. The estimated gaze frequency (Gf) and estimated gaze density (Gd) were used to represent the gaze behavior pattern of pedestrians, and the velocity variance (Vv), velocity direction (Vd), distance range (Dr) and mean distance (Dm) were used to represent the walking behavior pattern of pedestrians. The walking behavior patterns and gaze behavior patterns of real hidden followers and normal pedestrians are stitched into six-dimensional vectors, clipped into sequences of the same length, and used as positive and negative samples to train the time series classification model, and finally, the hidden follower recognition model is obtained.

IV. EXPERIMENTS

A. Dataset

In this paper, we use the Real-HFD dataset [9]. It is divided into videos of hidden following scenes and videos of non-hidden following scenes, containing a total of 160 minutes of 20 pedestrian videos, where the number of pedestrians in each video is 5 to 12. In the hidden following videos, each video contains 1 or 2 hidden follower-target pedestrian pairs and several unrelated passers-by. In the non-hidden following videos, each video contains 4 to 6 acquaintance pairs.

B. Compared Methods

In this paper, we compare traditional trajectory-based follow detection method [8], hidden follower detection method [9], and deep learning-based human behavior recognition methods [2], [3], [5].

C. Evaluation Metrics

Similar to [8], [9], we use the $F1$ score and *accuracy* to evaluate the performance of RHFD. For our problem, let $\mathcal{I} = \{I_1, I_2, \dots\}$ denote a set of time series with a hidden following relationship discovered by any method, $\mathcal{J} = \{J_1, J_2, \dots\}$ denote a set of time series with a non-hidden following relationship discovered by any method, with *Precision*, *Recall* and *Accuracy* defined as follows:

$$Precision = \frac{|\cup_{i,j} (I_i \cap I_j^*)|}{|\cup_{i,j} I_i|}, Recall = \frac{|\cup_{i,j} (I_i \cap I_j^*)|}{|\cup_{i,j} I_j^*|}, \quad (10)$$

$$Accuracy = \frac{|\cup_{i,j} (I_i \cap I_j^*)| + |\cup_{i,j} (J_i \cap J_j^*)|}{|\cup_{i,j} I_i| + |\cup_{i,j} J_i|}, \quad (11)$$

where $\mathcal{I}^* = \{I_1^*, I_2^*, \dots\}$ is the set of ground truth series of hidden following, $\mathcal{J}^* = \{J_1^*, J_2^*, \dots\}$ is the set of ground truth series of non-hidden following. Then, the $F1$ is

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}. \quad (12)$$

TABLE I
DIFFERENTIATION BETWEEN HIDDEN FOLLOWERS AND NORMAL
PEDESTRIANS MODES: P-VALUE.

	Gf	Gd	Vv	Vd
p	9.99E-16	1.92E-4	4.36E-160	1.23E-123

TABLE II
PERFORMANCE OF RECOGNIZING HIDDEN FOLLOWERS ON THE
REAL-HFD USING DIFFERENT MAPPING FUNCTIONS FOR THE ESTIMATED
GAZE STATE.(%)

Methods	Precision	Accuracy	F1	Recall
HFDF_G [9]	71.59	71.36	71.47	71.36
RHFD_G1	75.93	74.72	75.32	74.72
RHFD_G2	74.07	71.94	72.98	71.94
RHFD_G3	78.46	74.74	76.51	74.74
RHFD_G4	72.59	71.66	72.12	71.66

D. Experimental Results

1) *Difference Analysis*: We performed KS tests [14] on hidden followers and normal pedestrians in estimated gaze state frequency(Gf), estimated gaze state density(Gd), velocity variation(Vv), and velocity direction(Vd). When $p < 0.01$, it means that there is a significant difference between both. As shown in Table I, $p < 0.01$, which indicates a significant difference between hidden followers and normal pedestrians in both velocity modes and gaze modes.

2) *Comparisons with the State-of-the-arts*: Table II shows the performance of recognizing hidden followers on the Real-HFD using only the estimated gaze states under different mapping functions. The correspondence between the mapping function and model name is: y_1 corresponds to RHFD_G1, y_2 corresponds to RHFD_G2, y_3 corresponds to RHFD_G3, and y_4 corresponds to RHFD_G4. It can be seen that the use of a quadratic function, which converts the gaze direction to the estimated gaze state of the gaze behavior mode, is more helpful in recognizing hidden followers. As shown in Fig.3, the higher the estimated gaze state with the same gaze direction, the better the effect of hidden follower recognition, which indicates that the hidden followers rarely gaze at the target pedestrian horizontally when following the target pedestrian, but there exists a certain angle to hide their following intention. Our optimal method (RHFD_G3) improves *Accuracy* to 74.74% and *F1* to 76.51%. Compared with the state-of-the-art [9], the relative improvement in *Accuracy* is 4.7% and *F1* is 7.1%.

Table III shows the performance of collaborative gaze mode and spacing mode for recognizing hidden followers on Real-HFD. We use different mapping functions to convert the gaze direction of pedestrians into different gaze behavior patterns and construct four different collaborative models, where y_1 corresponds to RHFD_GS1, y_2 corresponds to RHFD_GS2, y_3 corresponds to RHFD_GS3, and y_4 corresponds to RHFD_GS4. It was found that the model with a collaborative gaze behavior pattern and the spacing pattern was more effective in identifying hidden followers when a quadratic function was used as the mapping function. In addition, we also found that

TABLE III
PERFORMANCE OF RECOGNIZING HIDDEN FOLLOWERS ON THE
REAL-HFD WITH DIFFERENT COLLABORATIVE SPACING MODE AND GAZE
MODE METHODS.(%)

Methods	Precision	Accuracy	F1	Recall
HFDF [9]	91.19	90.68	90.93	90.68
RHFD_GS1	95.01	94.72	94.86	94.72
RHFD_GS2	94.95	94.44	94.69	94.44
RHFD_GS3	95.18	95.00	95.09	95.00
RHFD_GS4	94.52	94.16	94.34	94.16

TABLE IV
PERFORMANCE OF RECOGNIZING HIDDEN FOLLOWERS ON THE
REAL-HFD USING DIFFERENT METHODS.(%)

Methods	Precision	Accuracy	F1	Recall
C3D [2]	53.30	49.91	50.51	46.03
I3D [3]	50.81	47.10	46.61	42.43
Slowfast [5]	57.63	53.71	53.33	48.52
Trajectory [8]	57.11	62.53	72.74	97.52
HFDF [9]	91.19	90.68	90.93	90.68
RHFD1	95.37	95.00	95.18	95.00
RHFD2	95.31	94.44	94.87	94.44
RHFD3	96.25	96.11	96.18	96.11
RHFD4	95.11	94.44	94.78	94.44

the larger the estimated gaze state was, the more beneficial the model combined with the spacing pattern was in identifying hidden followers for the same gaze direction. Our optimal method (RHFD_GS3) improves *Accuracy* to 95% and *F1* to 95.09%. The relative improvement of *Accuracy* is 4.8% and *F1* is 4.6% compared with the state-of-the-art method [9].

Table IV shows the performance of different methods to recognize hidden followers on the Real-HFD. We constructed four different models for the refined representation of gaze behavior and walking behavior by converting the gaze direction of pedestrians into different gaze behavior patterns. Specifically, these four models correspond to different mapping functions, i.e., y_1 corresponds to RHFD1, y_2 corresponds to RHFD2, y_3 corresponds to RHFD3, y_4 corresponds to RHFD4. Our optimal method (RHFD3, which is RHFD) improves the *Accuracy* to 96.11% and *F1* to 96.18%. The relative improvement in *Accuracy* is 104.1% and *F1* is 106.4% compared to deep learning-based human behavior recognition methods [2], [3], [5]. The relative improvement in *Accuracy* is 53.7% and *F1* is 32.2% compared to the traditional trajectory-based method [8]. Compared with the latest state-of-the-art [9], the *Accuracy* is relatively improved by 6%, and *F1* is relatively improved by 5.8%. RHFD outperforms the state-of-the-art methods in terms of *Accuracy*, *Precision*, and *F1*, which indicates that RHFD has significant advantages in recognizing hidden followers.

3) *Ablation Study*: We study the effectiveness of each part of our RHFD by analyzing the following baselines.

- RHFD_VS: To test the effect of the estimated gaze state on hidden follower recognition, we construct the RHFD_VS baseline by dropping the gaze mode module.
- RHFD_GS: We evaluate the necessity of pedestrian

TABLE V
PERFORMANCE OF RECOGNIZING HIDDEN FOLLOWERS AFTER REMOVING
DIFFERENT PARTS OF RHFD ON THE REAL-HFD.(%)

Methods	Precision	Accuracy	F1	Recall
RHFD_S	90.67	90.22	90.44	90.22
RHFD_GS	95.18	95.00	95.09	95.00
RHFD_VS	93.58	93.33	93.46	93.33
RHFD	96.25	96.11	96.18	96.11

walking velocity by comparing it with RHFD_GS. This baseline removes the velocity mode module from the RHFD framework.

- RHFD_S: To highlight the advantages of estimated gaze state and velocity variations of pedestrians, we recognize hidden followers by using only the spacing mode module.

After the ablation experiments, the experimental results are obtained as shown in Table V. It can be seen that the performance of these models is ordered according to $RHFD > RHFD_GS > RHFD_VS > RHFD_S$. Based on the observations and analysis, the following conclusions can be drawn:

- Both the estimated gaze state and pedestrian walking speed have positive effects on hidden follower identification, and their performance is better than using only the spacing model ($RHFD_GS > RHFD_S, RHFD_VS > RHFD_S$).
- The model combining the estimated gaze state and spacing pattern outperforms the model combining the pedestrian walking velocity and spacing pattern ($RHFD_GS > RHFD_VS$), which indicates that the estimated gaze state is more discriminative than the pedestrian walking velocity and can better distinguish the hidden followers from normal pedestrians.
- Combining the estimated gaze state and pedestrian walking velocity can improve the performance of the model ($RHFD > RHFD_GS, RHFD > RHFD_VS$), which indicates that the estimated gaze state and pedestrian walking velocity have complementary effects.

In summary, the results of the ablation experiments show that each part of the RHFD model plays a crucial role in the identification of hidden followers.

V. CONCLUSION

In this paper, we have proposed a Refined Hidden Follower Detection (RHFD) method to recognize hidden followers, which jointly uses estimated gaze state to refine the representation of complex pedestrian gaze behavior, using pedestrian velocity magnitude variation and direction to characterize the walking state of hidden followers in greater depth. We found that the gaze behavior of hidden followers has higher spatiotemporal focus and following, while normal pedestrians show more observation of their surroundings. Therefore, the proposed model can comprehensively represent the behavioral characteristics of hidden followers when following the target pedestrians. Evaluation of the surveillance video dataset shows

that RHFD achieves significant improvements over state-of-the-art methods. In the future, we will continue to extend RHFD to identify more types of pedestrian relationships and apply it to more complex scenarios, to further improve the accuracy and practicality of pedestrian relationship detection on foot.

ACKNOWLEDGMENT

The work is supported by the National Nature Science Foundation of China (No. U22A2035, U1803262, U1736206), National Social Science Fund of China (No. 19ZDA113), and the Application Foundation Frontier Project of Wuhan Science and Technology Bureau (No. 2020010601012288), and the Key Research and Development Program of Shaanxi under Program 2022KW-06.

REFERENCES

- [1] Karen Simonyan and Andrew Zisserman, "Two-stream convolutional networks for action recognition in videos," *Advances in neural information processing systems*, vol. 27, 2014.
- [2] Du Tran, Lubomir Bourdev, Rob Fergus, Lorenzo Torresani, and Manohar Paluri, "Learning spatiotemporal features with 3d convolutional networks," in *Proceedings of the IEEE international conference on computer vision*, 2015, pp. 4489–4497.
- [3] Joao Carreira and Andrew Zisserman, "Quo vadis, action recognition? a new model and the kinetics dataset," in *proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 6299–6308.
- [4] Swathikiran Sudhakaran, Sergio Escalera, and Oswald Lanz, "Lsta: Long short-term attention for egocentric action recognition," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 9954–9963.
- [5] Christoph Feichtenhofer, Haoqi Fan, Jitendra Malik, and Kaiming He, "Slowfast networks for video recognition," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2019, pp. 6202–6211.
- [6] Yi-Bin Cheng, Xipeng Chen, Junhong Chen, Pengxu Wei, Dongyu Zhang, and Liang Lin, "Hierarchical transformer: Unsupervised representation learning for skeleton-based human action recognition," in *2021 IEEE International Conference on Multimedia and Expo (ICME)*. IEEE, 2021, pp. 1–6.
- [7] Fernando de Lucca Siqueira and Vania Bogorny, "Discovering chasing behavior in moving object trajectories," *Transactions in GIS*, vol. 15, no. 5, pp. 667–688, 2011.
- [8] Zhenhui Li, Fei Wu, and Margaret C Crofoot, "Mining following relationships in movement data," in *2013 IEEE 13th International Conference on Data Mining*. IEEE, 2013, pp. 458–467.
- [9] Danni Xu, Ruimin Hu, Zheng Wang, Linbo Luo, Dengshi Li, and Wenjun Zeng, "Gaze-and spacing-flow unveil intentions: Hidden follower discovery," in *Proceedings of the 30th ACM International Conference on Multimedia*, 2022, pp. 2115–2123.
- [10] Danni Xu, Ruimin Hu, Zixiang Xiong, Zheng Wang, Linbo Luo, and Dengshi Li, "Trajectory is not enough: Hidden following detection," in *Proceedings of the 29th ACM International Conference on Multimedia*, 2021, pp. 5373–5381.
- [11] Jialian Wu, Jiale Cao, Liangchen Song, Yu Wang, Ming Yang, and Junsong Yuan, "Track to detect and segment: An online multi-object tracker," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2021, pp. 12352–12361.
- [12] Petr Kellnhofer, Adria Recasens, Simon Stent, Wojciech Matusik, and Antonio Torralba, "Gaze360: Physically unconstrained gaze estimation in the wild," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2019, pp. 6912–6921.
- [13] Preetiya Tanisaro and Gunther Heidemann, "Time series classification using time warping invariant echo state networks," in *2016 15th IEEE International Conference on Machine Learning and Applications (ICMLA)*. IEEE, 2016, pp. 831–836.
- [14] Vance W Berger and YanYan Zhou, "Kolmogorov-smirnov test: Overview," *Wiley statsref: Statistics reference online*, 2014.